

Workflow solutions

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BT acknowledges that workflow technology is now mature and ready for use as a core component of operational support systems. This paper focuses on the use of workflow as the central application service that interacts with new and legacy system applications, and supports users in managing their work.

Advances within BT have created a process implementation method that provides a common approach, a common development environment, and an operational environment. This method has now been used by several developments, with more under way.

The use of workflow technology in this way has the potential to reduce time to market, reduce development costs, and reduce operational costs, for medium-to-large-scale enterprise-wide developments throughout BT.

1. Introduction

This paper describes the current state-of-the-art in workflow technology used in BT. It describes how advances within BT have provided a common process-based implementation method, an infrastructure, and guidelines that both enhance and simplify workflow use.

Workflow technology [1] on its own provides many business benefits; however, within BT, workflow is integrated with databases, application (Web) servers, and business components, to extend the range of possible problems that can be addressed. This approach allows BT to increase the scope and usefulness of workflow technology.

This paper describes the method used, how the tools are used to translate and provide real solutions, and how a variety of solutions are now in use in BT. The paper concludes by looking at possible futures for the use of workflow technologies.

2. Process implementation method

In order to implement workflow-based solutions across business areas, capable of solving a variety of business issues, an approach or method is required. The process implementation method (PIM), described below, is used in this work.

The PIM provides the sequence of steps to:

- integrate disparate systems or applications via business services,
- provide the necessary interaction between users, and/or groups of users.

The PIM relies on an architectural framework provided by the workflow technology, and assumes system components and their infrastructure will be capable of providing any external (non-workflow) systems interaction.

The PIM prescribes that systems will be constructed from graphical process definitions. This allows requirements to be captured, often by end users using their own terminology. As a result all people performing all roles involved in the PIM life cycle gain a greater understanding as to what is required, and what will be developed and deployed.

Since the workflow engine interprets process definitions, simple systems can be easily and quickly constructed using the process definition alone. In practice, existing projects require reference data, integration to other systems, reports, etc. This means that simple systems are usually limited to only prototypes and demonstrators. However, since these simple systems can be built and deployed in a matter of hours, the PIM provides much needed clarification during the requirements capture stage of the life cycle.

More complex systems always require some form of traditional development, and for this the PIM assumes that component-based development (CBD) techniques will be used. In the PIM each CBD component becomes a step in a process definition. These steps are available from a library (catalogue) for use by process builders, and this provides, and enforces, a consistent approach.

CBD is too large an area to discuss in this paper, but BT has standardised on one approach to providing business services [2], and several external CBD references are available [3, 4].

Users of the rapid application development (RAD) [5] or dynamic systems development methods (DSDM) [6] should also consider the PIM. The PIM uses many of the underlying principals of RAD/DSDM, and provides workflow technology to facilitate the key stages. Use of the workflow process builder provides requirement capture and identifies components for design. Saving the model in the workflow engine delivers many of the traditional build-and-deployment capabilities. This leaves the actual component build as the remaining traditional development task. At all stages in the life cycle the model allows an iterative time box approach to be realised.

Use of the process implementation method, see Fig 1, implies that workflow based systems will:

- 1 layer on top of one or more traditional operational support systems (OSS), such as legacy systems, bought-in packaged systems, or other databases (access to these is obviously optional, but has to be catered for in the PIM),
- 2 interface to OSS, and perhaps more importantly interface to any other piece of automation/applications, through the use of components — these components are sometimes known as business services, or middleware [2], and scale in complexity from a simple database search, to providing integration to several OSS,
- 3 be driven by a workflow process model, constructed to determine the optimal route to provide an end-to-end solution to a problem — ancillary processes can also be used to interact with the main process to cater for things like errors, jeopardy handling, or periodic customer contact,

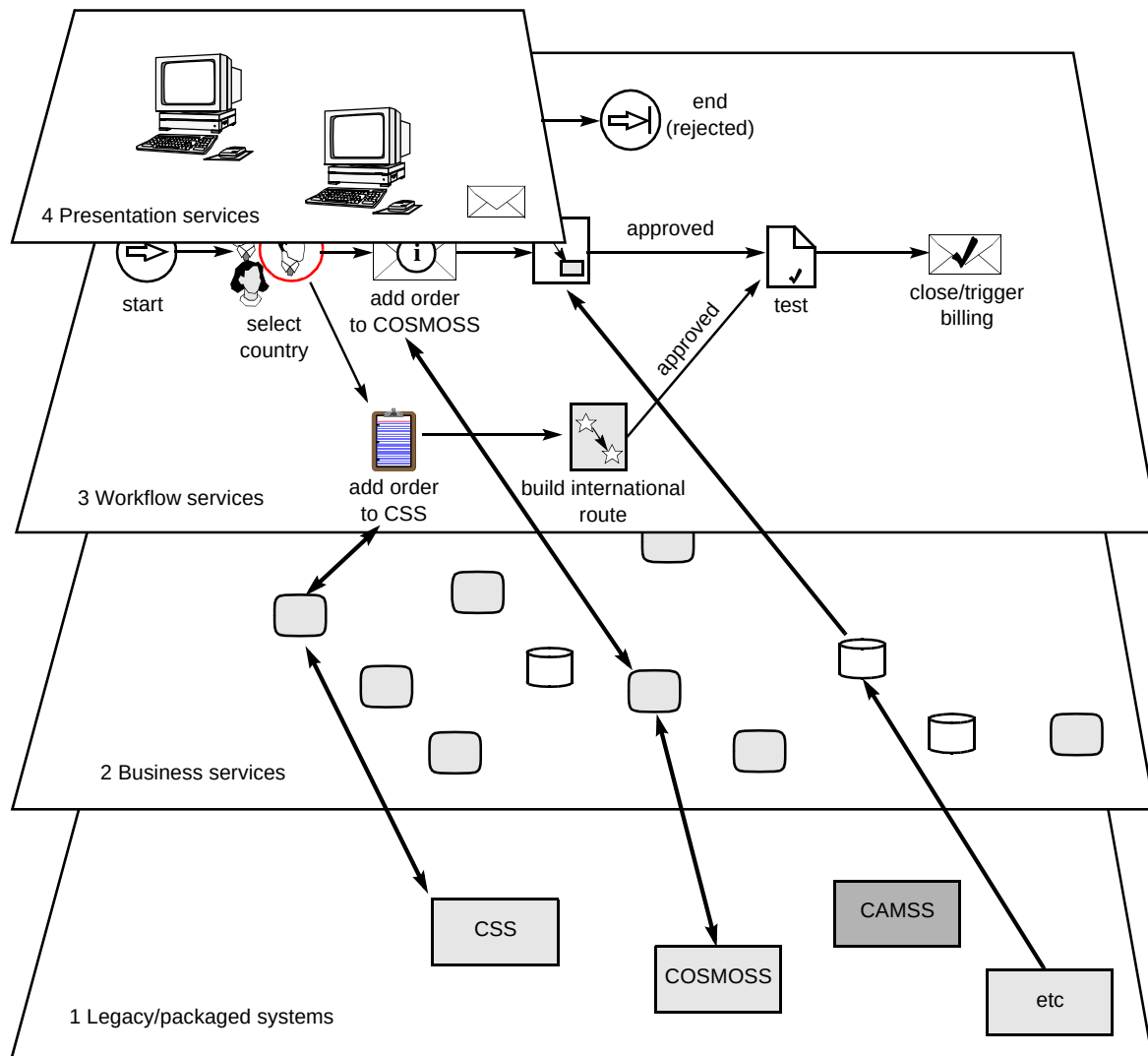


Fig 1 Process implementation method.

- 4 be capable of interacting with users in a variety of ways, predominantly via a WWW or e-mail interface — user steps are also described in the process model, allowing dynamic generation of the user interface at run time.

The method, shown in Fig 1, also implies that workflow should be capable of providing component integration in two ways.

- Synchronous

The first, and simpler approach, is that when a step in a process is activated the workflow system calls a component to automatically perform a piece of work. When the component has completed its task, control returns to the workflow engine. Workflow then interprets the model using the result from the component to determine what to do next. This is the traditional synchronous approach to development activities.

- Asynchronous

The second, and potentially more important approach, is that workflow systems can also work asynchronously. This is how users typically interact with workflow. In this case workflow decides what to do and who is to do it, then waits until the user eventually completes the task. The user response is actually a trigger back into the workflow engine causing it to start up again and to continue along the process definition. This asynchronous approach also works with components, to ensure that longer-lived components do not load or interfere with the capacity of the workflow engine to do other tasks. This

asynchronous approach is also used to provide a mechanism allowing workflow processes to be interrupted when external events occur.

3. Workflow solution designs

While the process implementation method provides the ideal approach on how to build solutions, workflow solutions designs must use available product sets and development tools to provide realistic systems.

BT uses its IT infrastructure programme to put in place a mandatory standard IT infrastructure with associated services and supporting business mechanisms. The standard IT infrastructure is the set of products and services that BT uses to run its internal computing operations. This recommends the use of Oracle products to implement workflow solution designs, although the PIM could be applied to other leading workflow suppliers outside BT.

When a solution design is looked at in its entirety, key factors such as the components shown in Fig 2, make it obvious that workflow alone does not provide the full solution. However, workflow is aligned with traditional Oracle development products that do allow components, and therefore a complete solution, to be developed in a consistent manner.

Oracle Workflow also comprises an Oracle database and an Oracle application (WWW) server. It is the use of the three products together that provides a greater benefit than if each were used individually. The workflow engine is code resident in the Oracle database, which means that database integration is extremely easy and efficient.

To implement workflow solutions using the process implementation method three key factors, as shown in Fig 2

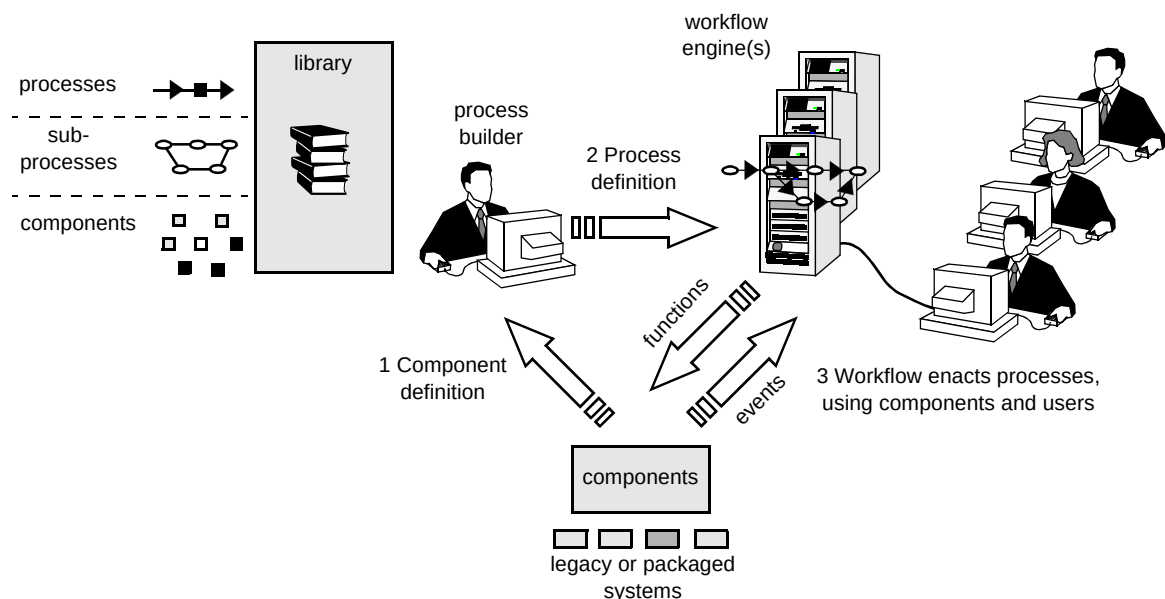


Fig 2 Implementation steps.

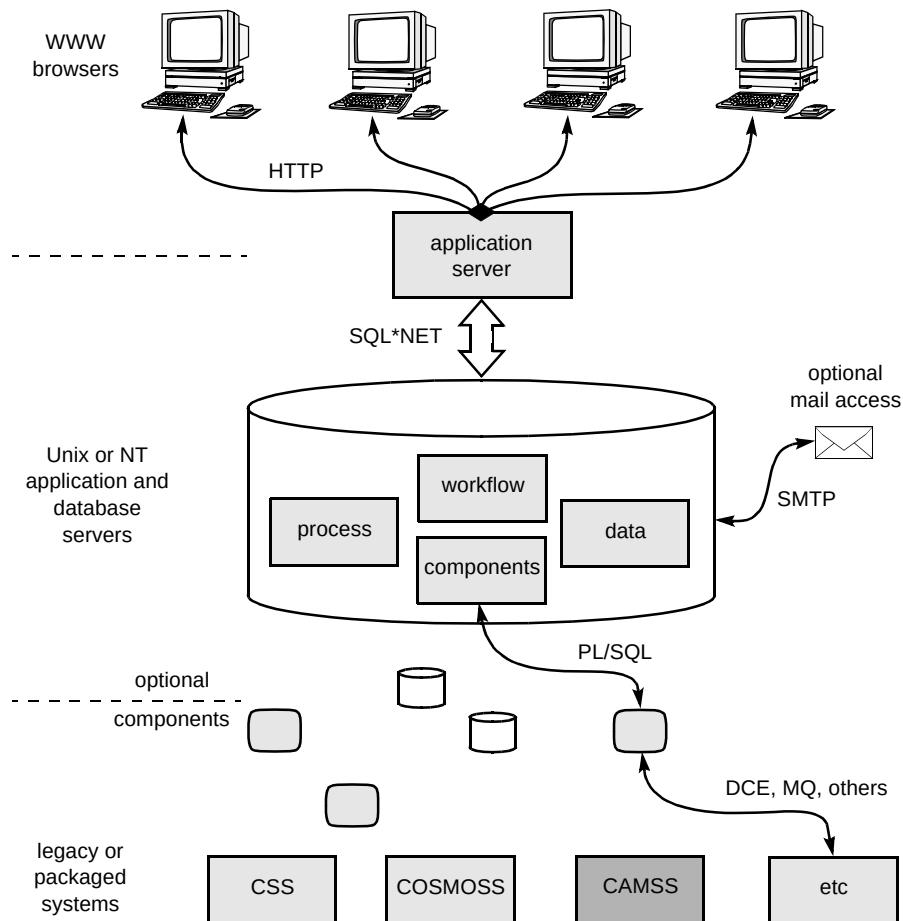


Fig 3 Workflow solution deployment.

and described below, are needed. This leaves deployment, as shown in Fig 3, to turn the solution into a usable system.

3.1 Components

The first area that has to be considered relates to components, specifically their build; component build remains a traditional development task.

While workflow triggers a component into use, it provides only a standard API, a description of what it has to do, and key data, to allow them to be created.

Components in Oracle Workflow are actually PL/SQL procedures resident in the database. More complex components need to get out of the database (and Oracle provide a variety of options for this) in order to interact with, or be interrupted by, external systems.

External components have been built using a variety of programming techniques, ranging from VB on NT, or C++ on Unix. These components in turn access legacy systems in a variety of ways, the VB components driving fast-track transaction services [7], and the C++ components using DCE or MQ Middleware [2].

In initial projects using the PIM, the components were described as the initial processes were built. While the components were being built, workflow tasks users to perform the activities manually, thus allowing initial deployment earlier. The manual tasks will seamlessly be replaced when the components arrive from developers.

Now that these initial projects have provided the learning experience, common components are starting to emerge, and are available for use in new projects.

One hidden advantage of this CBD approach is that the overall design problem is broken into smaller isolated areas of development; each component has to be capable of independent/isolated operation when workflow determines it has to be invoked. This approach helps simplify the overall development.

3.2 Processes

Components provided by developers are stored in library files for use by process builders. They use these to construct processes, interspersing them with user steps, to achieve the desired goals.

User steps are defined in the process builder at this stage, and the workflow engine uses their definition at run time to dynamically generate user interfaces via WWW or mail.

Building workflow solutions typically involves capturing processes from workshops comprising end users. From these models multiple components emerge (or are already available) that can be developed, initially as a stub and eventually as the functioning code. During the development of the components the model is usually re-engineered to add or correct user tasks, often by running orders/jobs through the processes on the development system.

It should be noted that the PIM also supports user processes that consist of components sequenced by workflow, or consist of processes that only comprise user tasks sequenced by workflow.

Finally, in the same way that models can be built from components and/or user tasks, models can also be built from other models. Oracle is completely hierarchical in this approach allowing process build to be segmented.

3.3 Configuration

Once processes have been developed they have to be deployed, or enacted. This is as simple as saving the model into the database where the run-time side of workflow interprets the process model and pushes work through the system, interacting with components, or users, as required.

This obviously assumes the components have been built and component build remains the development area of the overall solution.

3.4 Deployment

Most workflow solutions in BT use intranet WWW access via a standard browser, allowing users to interact with the system, as shown earlier in Fig 3.

WWW interfaces are recommended since they are built dynamically, direct from the process model definition, and are easily supported by existing browsers, although mail support is also provided. WWW access is also recommended for users whose work delivery is guided by the workflow system. A few solutions also use e-mail, as this is suitable for users who infrequently interact with the system.

An Oracle application (Web) server accesses the database to provide WWW access. The database contains the workflow engine, process definitions, instance (run-time) data, and PL/SQL components.

When a process reaches an automated step it invokes a component. Although Fig 3 shows these components as being outside the database, in practice all start and/or end in the database. In the simpler components such as data table access, or data-based logic algorithms, the complete component resides in a database package. In complex components, such as those that connect to an external application, the component interacts with a component layer (or infrastructure) outside the database that actually performs the access to the other applications.

4. BT enhanced workflow

Commercial products rarely supply everything that is needed to build BT solutions without some additional effort, and workflow products are no exception.

However, the IT Infrastructure Program has guided development activities to provide a common infrastructure layer, reference designs, and general workflow guidelines, to simplify future developments.

BT Additions range from a BT 'look and feel' for user interfaces (leveraging capabilities within the application server), to more important enhancements such as an enhanced user worklist [8] (leveraging capabilities within the workflow engine). Both of these examples are shown by the screen-shot in Fig 4.

This 'wrapping' of workflow extends to the provision of functionality for consistent event handling for external synchronous, asynchronous, and even manual interfaces.

Guidelines [8] are also available to help new projects avoid common pitfalls, and to share the experiences of past developments. The guidelines include topics such as naming standards, process model consistency, workflow and external system applications interfaces, Microsoft Word document generation, etc.

Finally, a reference design is available that can be used as a template for new projects. All workflow-centric solutions that use the PIM and infrastructure inherit a framework on which to build consistent solutions.

This approach also helps in one of the largest problem areas in BT, that of systems interaction. Workflow solutions currently have interfaces into legacy systems via fast-track transaction services [9], or middleware for distributed systems [2], and also support messaging.

5. Workflow solutions in BT

In the last year BT started several workflow projects, and more have started development for release this year. Existing projects cover a variety of different business areas across BT, have a variety of different uses, and are being developed in operational and engineering centres throughout the UK.

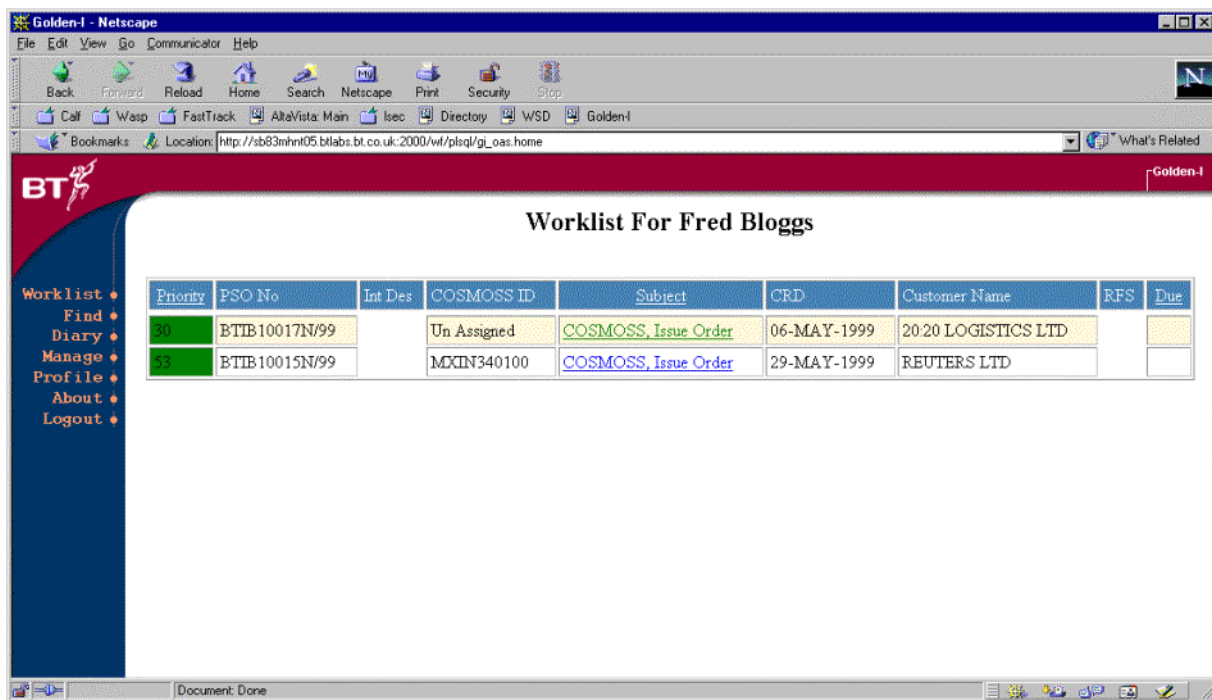


Fig 4 BT enhanced work list.

The business drivers in all projects involved the following to a varying degree — reduced time to market, reduced operational costs, improved operational quality, reduced development costs, and the use of a process that is known and consistent.

Perhaps the most straightforward project involves ordering equipment, and is intended for use within the BT Intranet. Although it was not the first project to start, it was the quickest project to be designed and developed, and the first to be deployed. WWW-based order forms are used to invoke a workflow process that routes and manages the work through to completion. A spin-off project is now being investigated for use externally, allowing customers to order simple products from the Internet.

A separate project is now available on the Internet, allowing other telecommunications providers to place orders on to a BT provisioning system. A workflow process is started for each order to place order lines on other order provisioning systems, and to guide BT resources to oversee the order delivery.

Perhaps the most complex project, in terms of process and component developments, is the control system for all international private circuit provisions. From a workflow point of view this has some 30 top-level processes, made up of many more constituent sub-processes. Each process comprises activities available from a catalogue of some 200 possible user steps and 100 automatic components. Workflow was viewed as the natural development approach given the complexity of the processes involved, and has provided a common viewpoint throughout the design and development.

The final solution described here uses the same workflow engine, and the same component-based approach, but the workflow engine actually forms part of the Oracle financial applications suite. In this example the workflow engine was provided with additional processes. The new processes allow users to enter purchase requisitions, which then go through an approval process. Components are used to access the finance database for queries such as approval limits, management hierarchy, etc.

All of the projects share several common concepts. They all need reference data, need to store and retain order data, need accurate management of the work flowing through the process, need to access other systems or applications, and need easy deployment for users across the UK. In two projects user access is needed worldwide, in another two access is provided for BT customers. All projects had the concept that a business process needed to be followed. Its definition became the starting point of the project, and its definition remains as the running system.

A full list of projects and contacts are available from the workflow solution design WWW site.

6. Futures

Obviously workflow technologies continue to improve. A new release of Oracle Workflow is available providing many enhancements, for example, alternate system integration through message-based approaches built from the process definition.

This will be of great help with event-based systems integration now available on legacy applications.

Needless to say the workflow market-place continues to evolve and BT continues to monitor these advances to leverage further benefits from workflow.

The emergence of mobile-based communications appliances, and the use of PDA types of device, will allow a greater range of deployment options for workflow-based solutions. This will also cater for connected and mobile (roving) approaches.

Simplification of the middleware area using a standard approach will reduce the number of component integration types that workflow has to cater for. BT is currently defining new standards for a common middleware, and tests with workflow should be complete this autumn.

As BT's experience in workflow grows, and as more projects are delivered, the number of components available for use in workflow processes also grows. Although this is currently unco-ordinated, the concept of standard libraries exists and some projects are sharing BT-specific libraries layered above Oracle's own standard libraries. Work will take place this year to develop a component library approach throughout BT allowing new projects to exploit existing developments.

The use of agents to enhance workflow [10] will allow the use of algorithms to provide project-specific features, for example alternate role assignment, load balancing, perhaps even solutions where processes are created, used, and destroyed, dynamically. Work is being planned to move this important area forward by putting hooks into workflow to allow interaction with agents. An initial trial is planned allowing agents to predict deliveries based on knowledge of previous process flows.

Knowledge of which projects are suitable for workflow solutions design is currently a skilled area, but will become easier when work on suitability filters is completed later this year.

7. Conclusions

BT's exploitation of the PIM and workflow technology depends on a standard approach that uses a standard infrastructure that in turn uses standard components. This can only be successful if it is used with the knowledge of where workflow solutions should be applied.

This paper has shown how the use of the PIM allows a process definition to dynamically create workflow-based solutions that reduce traditional development times and increase overall flexibility. Used in this way workflow provides a consistent link from requirements capture all the way to operational support.

It has also shown how workflow, databases, and application servers, layered on component-based devel-

opments, are linked together in a common development approach to provide a diverse range of solutions in BT, and has provided a summary of the recent workflow solutions in use in BT across a variety of business areas, demonstrating how each needed quick, flexible, and ultimately cost-effective solutions for data and WWW-centric application areas.

Within BT, workflow solutions provide another development facet to most Oracle database and/or Oracle application (Web) server solutions.

While it would be useful to quantify these statements in practice, it cannot be done yet. However, there is now a sharing of the workflow infrastructure between projects, and BT has experience of component reuse that is now starting to happen with workflow components as well, both of which will reduce future development times and costs. Flexibility is also apparent with new processes being added, and some existing processes being re-engineered successfully.

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